UNIT-I

1. **Define Graphics:**

Graphics are visual presentations on some surface such as wall, canvas, computer screen or paper. Graphics can be functional, artistic, realistic or imaginary. E.g: Photographs, drawings, line art, graphs, diagrams, symbols, maps, geometric designs & engineering drawings.

2. **Uses of Graphics**

Graphics are often used to point readers and viewers to particular information.The Practical areas where graphics are applied are given below:

a) Business
b) Advertising
c) Political News
d) Education
e) Film & Animation
f) Graphics Education

3. **Define pixel.**

A pixel is generally thought of as the smallest complete sample of an image.

(or)

A pixel is a single point in a graphic image.

- Each such point (or) information element is not really a dot, nor a square.
- Pixels are measured in dpi (dots per inch) (or) ppi (pixels per inch).

4. **Define Resolution.**

The number of pixels in an image is called the Resolution.

Eg: 640 by 480 display

(or)

640* 480 display

ie. 640 pixels from side to side & 480 pixels from top to bottom

640 * 480 = 307,200 pixels

(Or)

0.3 Megapixels.

5. Define Bit per pixel:

- The number of distinct colors that can be represented by a pixel depends on the number of “bits per pixel” (bpp).
- The maximum number of colors a pixel can take can be found by taking two to the power of the color depth.

Eg: 256 colors = 2^8, 8 bpp.

6. Define Sub Pixel

- Many display and image acquisition systems are not capable of displaying (or) sensing the different color channels at the same site.
- The above problem is generally resolved by using multiple sub pixels, each of which handles a single color channel either Red, Green (or) Blue.

Example,

i) LCD’s typically dividing each pixel horizontally into three sub pixels.

ii) Most LCD displays divide each pixel into four sub pixels; one Red, one Green and two Blue.

7. Define Mega pixel.

A megapixel is 1 million pixels.

Eg: 2048 * 1536 pixels = 3.1 megapixel (3,145,728 pixels).

8. Define Spatial resolution.

The measure of how closely lines can be resolved in an image is called spatial resolution, and it depends on properties of the system creating the image, not just the pixel resolution in pixels per inch (ppi).

9. Define Display resolution.

The display resolution of a digital television or display device is the number of distinct pixels in each dimension that can be displayed.

10. Define Video display devices and explain its types.

- The primary output device in a graphics system is a video monitor.
- The operation of most video monitors is based on the standard cathode-ray tube.

The video display devices discussed here are given below:

- Refresh cathode ray tubes
- Raster scan displays
- Random scan displays
- Color CRT monitors
- Direct view storage tubes
- Flat panel displays &
- Three dimensional viewing devices
- Refresh Cathode Ray Tubes.

11. What are the two basic techniques used for producing color display with CRT?

The two basic techniques are used for producing color display with a CRT namely

- Beam penetration method.
- Shadow mask method.

12. What are the different color combinations?

The 3 different color combinations are:

- Yellow – green & red beams
- Magenta - blue & red beams
- Cyan – blue & green beams

13. List the advantages and disadvantages of Direct View Storage Tubes (DVST).

Advantages:

i) No refreshing is needed.

ii) Very complex pictures can be displayed at very high resolutions without flicker.

Disadvantages:

i) Selected parts of a picture cannot be erased.

ii) They ordinarily do not display color.

14. List the uses of flat panel displays.

Uses of flat panel displays are:

- TV monitors
- Calculators
- Pocket video games
- Laptop computers
- Armrest viewing of movies on airplanes.
- Advertisement board in elevators.

15. List the categories of Flat Panel Displays.

There are two major categories of flat panel displays

- Emissive displays
- Non emissive displays

16. State the examples of Emissive Displays.

The few examples for Emissive are:

i) Plasma panels

ii) Thin film electroluminescent displays

iii) Light emitting diodes.

17. Explain Non emissive displays

These displays use optical effects to convert sunlight from some other source into graphics patterns.

Examples: Liquid crystal device.

18. Define Liquid Crystal Device.(may 2012)

- These displays are commonly used in small systems such as calculators & portable laptop computers.
- The term liquid crystal refers to compounds which have a crystalline arrangement of molecules. However it flows like liquid.

19. Give some real – time example for 3D Viewing Devices.

- Geisco space graph system – used in medical applications
- For analyzing data from ultrasonography & CAT scan devices.
- In geological applications to analyze topographical & scismic data.

20. Define Input devices and list various input devices.

Input Devices:

- Various devices are available for data input on graphics workstations.
- Most systems have a keyboard & one or more additional devices specially designed for interactive input.

The various input devices are:

i) keyboards
ii) mouse
iii) trackball & space ball
iv) joysticks
v) data glove
vi) digitizers
vii) image scanners
viii) touch panels
ix) light pens
x) voice systems.

21. Explain the functions of keyboards

The keyboard is an efficient device for inputting such nongraphic data as picture labels associated with a graphic display.

The common features on general purpose keyboards are

i) cursor-control keys

ii) function key

22. What is Trackball?

- Trackball is a ball that can be rotated with the fingers or palm of the hand to produce screen cursor movements.
- Potentiometers attached to the ball measure the amount & direction of rotation.
- They are often mounted on keyboard or z-mouse.

23. What is Spaceball?

- A space ball provides size degrees of freedom.
- A space ball does not actually move. Strain gauges measure the amount of pressure applied to the space ball.

Applications where space balls are used are

i) 3D positioning

ii) Virtual reality systems

iii) Modeling

iv) Animation

v) CAD

24. What are Joysticks?

- A joystick consists of a small vertical lever called the stick
- The stick is mounted on a base which steers the screen around.

There are 2 types of joystick

i) movable stick joystick

ii) non-movable stick joystick

25. What is a movable Stick Joystick?

- The stick is actually placed in the center position.
- Screen cursor movement is achieved by moving the stick in any direction from the center.
- Potentiometers also mounted at the base of the joystick & they measure the amount of movement.
- Potentiometers also return the stick to the center position when released.
- In another type of movable joystick the stick is used to activate switches that cause the screen cursor to move at a constant rate in the selected direction.

26. What is a Non Movable Stick Joystick?

- Non Movable Stick joysticks are also called isometric joysticks.
- They have a non movable stick & pressure applied on the joystick is measured by strain gauges & converted to the movement of the cursor in the specified direction.

27. What are Digitizers?

- A digitizer is a common device for drawing, painting or interactively selecting co-ordinating positions on an object.
- Digitizers are used to input co-ordinate values in either a 2D or 3D space.
- A digitizer is used to scan over a drawing or object & to input a set of discrete co-ordinate positions.
- Graphics tablet is an example of digitizer.

28. What are Image Scanners?

- An image scanner is used for storing drawings, graph, color & black and white photos or text by passing an optical scanning mechanism for computer processing.
- The gradations of gray scale or color are then recorded and stored in an array.
- Transformations can be applied to rotate, scale or crop the picture to a particular screen area.
- Various image processing methods can be applied to modify the array representations or text & they come in a variety of sizes & capabilities.

29. What are Touch Panels?

- Touch panel allow displayed objects or screen positions to be selected with the touch of a finger.
- A typical application of touch panel's is for the selection of processing options that are represented with graphical icons.
    - Eg ATM center touch panel

30. What is a Light Pen?

- Light pens are pencil-shaped devices used to select screen positions by detecting the light coming from points on the CRT screen.
- The recorded light pen co-ordinates can be used to position an object or to select a processing option.

31. List the disadvantages of light pen.

i) Screen image obscured by hand & light pen
ii) Prolonged use causes arm fatigue
iii) Light pens require special implementation for some applications.
iv) Sometimes light pens give false readings due to background lighting in room.

32. List the names of few Output Devices.

Few available output devices are:

- Speakers
- Microphone
- Printer

33. What are Speakers?

Speakers are the primary method of sound output in most computers today. Sound is translated from bits to electrical signals in a sound card, which then channels the signals to the speakers.

34. What is a Microphone?

A microphone is a device used to change sound into electric signals. Microphones are used in telephones, tape recorders, hearing aids and many other devices.

35. What are printers and name its types?

- A printer is a device that prints text or images, making a physical copy (usually with some kind of ink on paper).

Printers are classified into two types:

Impact and non-impact printers**.**

36. What are Impact and non-impact printers?

(i) Impact printers rely upon a mechanical impact to transfer ink to paper. Early printers were more like electric typewriters, striking an ink ribbon against the paper with a lever with a raised image of a letter on the end, or a rotating ball with the same.

(ii) Non-impact printers depend on other ways of creating an image. Plotters, covered separately, can also be used as printers.

37. What are Inkjet printers?

An inkjet printer is a type of computer printer that creates a digital image by propelling variably-sized droplets of liquid material (ink) onto a page. Inkjet printers are the most common type of printer and range from small inexpensive consumer models to very large and expensive professional machines.

38. What are hard-copy Devices?

- Hard copy output for images can be obtained in several formats
- Users can put pictures on paper by directing graphics output to a printer or plotter.
- The quality of pictures obtained from a device depends on the dot size & dot per inch or lines per inch.

39. What is a Dot Matrix Printer?

- It is an example of impact printer
- Dot matrix printers have a dot matrix print head containing a rectangular array of protruding pins

- The no. of pins depends on the quality of the printer
- Individual characters or graphic patterns are obtained by retracting certain pins so that the remaining pins form the pattern to be printed

40. What is an Electrostatic Printer?

- These printers place a negative charge on the paper one complete row at a time along the length of the paper.
- The paper is then exposed to the toner
- The toner is positively charged and is attracted to negative charged areas on the paper.

42. Define Graphics Kernel System.

- GKS defines a basic two-dimensional graphics system with: uniform input and output primitives; a uniform interface to and from a GKS metafile for storing and transferring graphics information.
- It supports a wide range of graphics output devices including such as printers, plotters, vector graphics devices, storage tubes, refresh displays, raster displays, and microfilm recorders.

11 marks:

1.List out and discuss the video display devices.(apr 2013)(apr 2012)

Video Display Devices

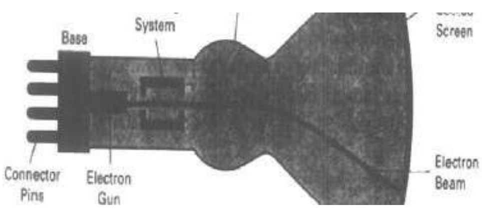
- The primary output device in a graphics system is a video monitor.
- The operation of most video monitors is based on the standard cathode-ray tube.

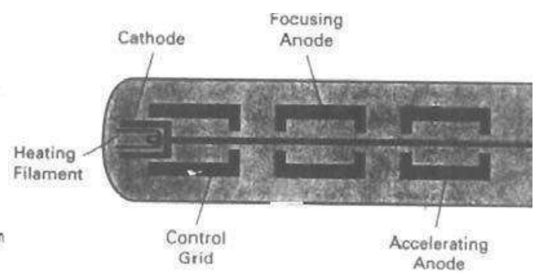
The video display devices discussed here are given below:

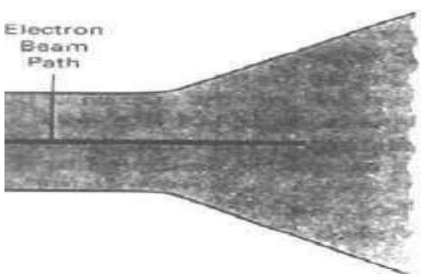
- Refresh cathode ray tubes
- Raster scan displays
- Random scan displays
- Color CRT monitors
- Direct view storage tubes
- Flat panel displays &
- Three dimensional viewing devices
- Refresh Cathode Ray Tube

- A beam of electrons emitted by an electron gun passes through focusing & deflecting systems that direct the beam toward specified positions on the phosphor-coated screen.
- The phosphor then emit a small spot of light at each position contacted by the electron beam. Because the light emitted by the phosphor fades very rapidly some method is needed for maintaining the screen picture.
- One way is to redraw the picture repeatedly by quickly directing the electron beam back over the same points.

This type of display is called a refresh CRT.





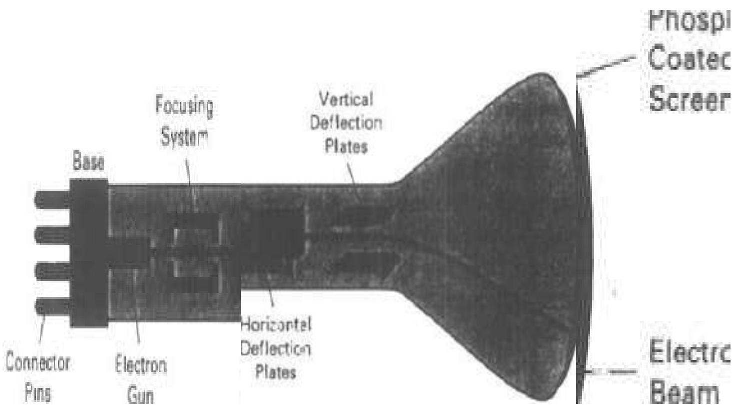


Working

- The primary components of an electron gun are heated metal cathode and a control grid.
- Heat is supplied to the cathode by directing a current through a coil of wire called the filament inside the cylindrical cathode structure.
- This causes electrons to be boiled off
- In the vaccum inside the CRT envelope the free negatively charged electrons are then accelerated toward the phosphor coating by a high positive voltage.
- The accelerating voltage can be generated with a Positively charged metal coating on the inside of the CRT near the phosphor screen.

Electrostatic Deflection

- Deflection of the electron beam can be controlled either with electric fields or with magnetic fields.
- Cathode ray tubes constructed with magnetic deflection coils mounted on the outside of the CRT envelope.
- Two pairs of coils are used.



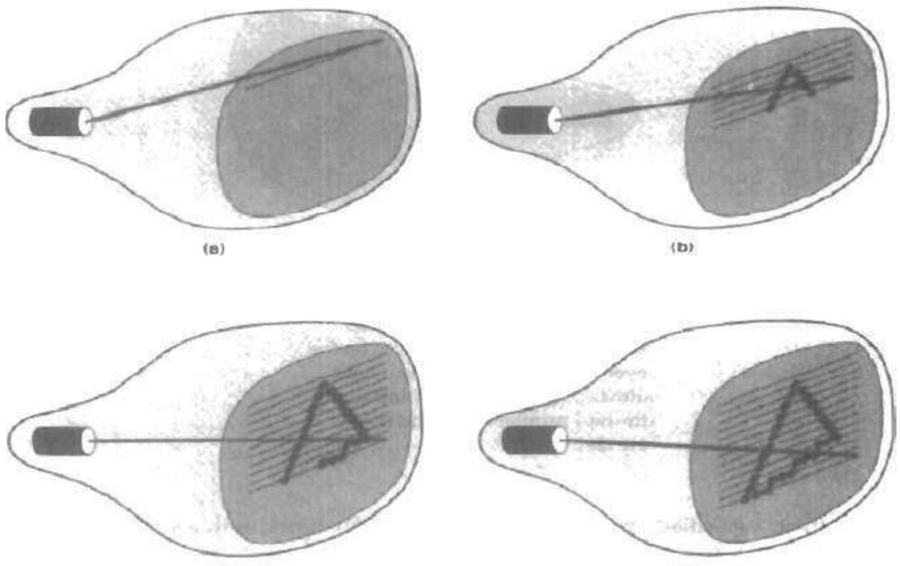
- One pair is mounted on the top & bottom of the neck.
- Other pair is mounted on the opposite sides of the neck.
- The magnetic field produced by each pair is of coils results in transverse deflection force that is perpendicular both to the direction of the magnetic field & to the direction of travel of the electron beam.

Raster Scan Displays

The most common type of graphics monitor employing a CRT is the Raster scan display.

Working Principle

- In a raster scan system the electron beam is swept across the screen one row at a time from top to bottom.
- When the electron beam moves across each row the beam intensity is turned on and off to create a pattern of illuminated spots.



- Picture definition is stored in memory area called Refresh buffer (or) Frame buffer.
- The frame buffer holds the set of intensity values for all the screen points.
- The stored intensity values are then retrieved from the refresh buffer and painted on the screen one row at a time.
- One row is also referred to as Scan Line.
- Each screen point is referred to as Pixel

Bit per Pixel

- Intensity range for pixel positions depend on the capability of the raster system.
- In black & white system each screen point is either on (or) off.

- Only one bit per pixel is needed to control the intensity of screen positions.

1 – electron beam is turned on

0 – electron beam is turned off

- High quality systems use upto 24 bits per pixel. The frame buffer of such system require several MB's of storage for the frame buffer.

Refreshing

- Refreshing of Raster scan displays is carried out as the rate of 60 to 80 frames per second.
- Refresh rates are described in units of cycles per second or Hertz.

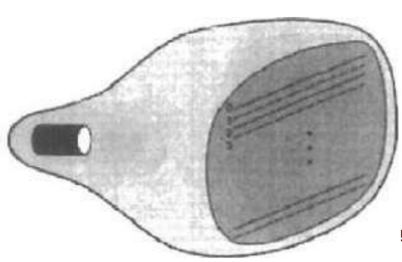
  1 cycle – 1 frame

- At the end of each scan line the electron beam returns to the left side of the screen to begin displaying the next scan line.
- The return of the electron beam to the left of the screen is called the horizontal retrace.
- At the end of each frame the electron beam returns to the top left corner of the screen.
- This is referred to as vertical retrace.

- **Random – Scan Systems**

- In a Random- scan display unit a CRT has the electron beam directed only to the parts of the screen where a picture is to be drawn.
- Random scan monitors draw a picture one line at a time & for this reason they are referred to as vector displays.

The component lines of a picture can be drawn and refreshed by as random scan system in any specified order.



Refresh Rate

- Refresh rate depends on the number of lines to be displayed.
- A refresh buffer is used to store picture definition. The line drawing commands are stored in the refresh buffer.
- To display a specified picture the system processes the set of drawing commands.
- Random scan displays are designed to draw all the components lines of a picture 30 to 60 times each second.
- Random scan system are designed for line drawing applications and cannot display realistic shaded scenes.
- Random scan displays produce smooth line drawings because the CRT beam directly follows the line path.

- **Color CRT Monitors**

A CRT monitor displays color pictures by using a combination of phosphors that emit different colored light.

Two basic techniques are used for producing color display with a CRT namely

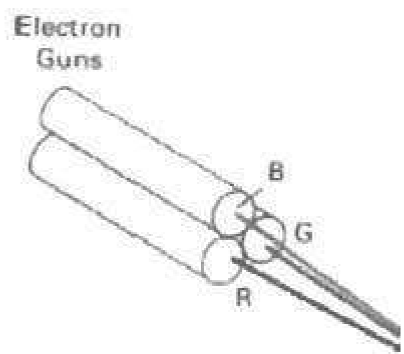
i) Beam penetration method.
ii) Shadow mask method.

Beam Penetration Method

- This method is used with random scan monitors.
- Here two layers of phosphor namely red & green are coated onto the inside of the CRT screen.

The displayed color depends on how far the electron beam penetrates into the phosphor layers.

- A slow beam of electrons penetrates through the screen & excites only the outer red layer.
- A beam of very fast electrons penetrates through the red layer and excites the inner green layer.
- An intermediate beam speed produces orange & yellow color (combinations of red and green lights)
- The speed of the electrons & hence the screen color at any point is controlled by the beam acceleration voltage.



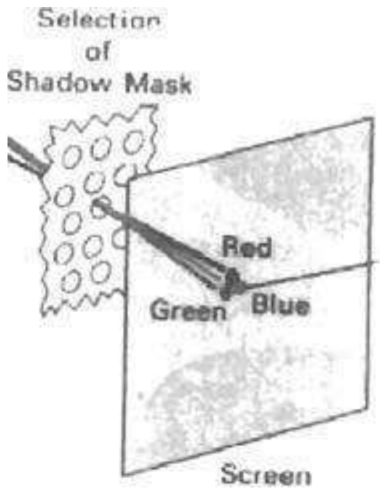
Shadow Mask Methods

➢ This method is commonly used in Raster scan systems.

A shadow mask CRT has three phosphor color dots at each pixel position.

- One for red light, one for green light & one for blue light.

➢ This type of CRT has three electron guns, one for each color dot.
➢ This CRT also has a shadow mask grid just behind the phosphor coated screen.



- The three electron beams are deflected & focused as a group onto the shadow mask.
- The shadow mask are aligned with the phosphor dot patterns.
- When the beams pass through the holes in the shadow mask they activate a dot triangle.
- Another configuration for the electron guns is an in-line arrangement where the 3 electron beams are aligned to a single scan line.
- By varying the intensity of levels of the three electron beams various color variations are obtained.

The color got depends on the amount of red, green and blue phosphors.

- A white area indicates that all the three electron beams are with the same intensity.

Color Combinations

Yellow – green & red beams

Magenta - blue & red beams

Cyan – blue & green beams

More sophisticated systems can set intermediate intensity levels for the electron beams.

- **DIRECT VIEW STORAGE TUBES**

An alternative method for maintaining a screen image is to store the picture information inside the CRT instead of refreshing the screen.

Two electron guns are used in DVST

DVST stores the picture information as a change distribution just behind the phosphor coated screen.

The primary electron gun stores the picture pattern.

The flood gun maintains the picture display.

Advantages

iii) No refreshing is needed.
iv) Very complex pictures can be displayed at very high resolutions without flicker.

Disadvantages

iii) Selected parts of a picture cannot be erased.
iv) They ordinarily do not display color.

Flat Panel Display

Introduction

The term refers to a class of video devices that have reduced volume, weight & power requirements.

- A significant feature of that panel displays is that they are thinner than CRT’s.
- Flat panel displays are available as pocket notepads.

Uses of flat panel displays are

- TV monitors
- Calculators
- Pocket video games
- Laptop computers
- Armrest viewing of movies on airplanes.
- Advertisement board in elevators

Categories

There are two major categories of flat panel displays

- Emissive displays
- Non emissive displays
- Emissive displays

Convert electrical energy into light.

Examples

iv) Plasma panels
v) Thin film electroluminescent displays
vi) Light emitting diodes.

Plasma Panels

- Plasma Panels Also called as Gas-discharge displays.
- Plasma panels are constructed by filling the region between two glass plates with a mixture of gases that usually include Neon.
- A series of vertical plates is placed in one glass panel.
- A series of horizontal plates is built into the other glass panel.
- When firing voltages are applied to a pair of horizontal & vertical conductors the gas at the intersection of the plates breaks down into glowing plasma of electrons & ions.

- Alternating current AC methods are used to provide faster application of the firing voltages & thus brighter displays.
- One advantage of plasma panels was that they were monochromatic devices but systems have been developed that are now capable of displaying color & grayscale.

Thin Film Electromagnetic Displays

- These displays are similar in construction to a plasma panel.
- The difference is that the region between the glass plates is filled with a phosphor, such as Zinc Sulphide doped with manganese instead of a gas.
- When a high voltage is applied to a pair of crossing electrodes the phosphor becomes a conductor in the area of intersection of the electrodes.
- The manganese atoms absorb the electrical energy as a spot of light.
- Electroluminescent displays require more power than plasma panels.
- Good color & gray scale displays are hard to achieve.

Light Emitting Diode (LED)

- A matrix of diodes is arranged to form the pixel positions in the display.
- Picture definition is stored in a refresh buffer.
- Information is read from the refresh buffer & converted to voltage levels that are applied to the diodes to produce the light patterns in the display.

i) Non emissive displays

These displays use optical effects to convert sunlight from some other source into graphics patterns.

Examples

Liquid crystal device.

- **Liquid Crystal Device (LCD's)**

- These displays are commonly used in small systems such as calculators & portable laptop computers.
- The term liquid crystal refers to compounds which have a crystalline arrangement of molecules. However it flows like liquid.
- Flat panel displays use nematic (thread like) liquid crystal compounds.
- The liquid crystal material is sandwiched between two glass plates each containing a light polarizer at right angles to the plate.
- Horizontal conductors are built into one glass plate & vertical conductors are built into another glass plate.
- The intersection of the two conductors define a pixel position.
- Polarized light passing through the material is twisted so that it will pass through the opposite polarizer.
- The light is then reflected back to the viewer.
- This type of flat panel device is referred to as a passive matrix LCD.
- Another method for constructing LCD’s is to place te transistor at each pixel location using thin- film transistor technology. These devices are called Active – Matrix LCD.

Three Dimensional Viewing Devices

- Graphics monitors for the display of three dimensional scenes have been devised using a technique that reflects a CRT image from a vibrating, flexible mirror.
- As the mirror vibrates it changes focal length.
- These vibrations are synchronized with the display of on a CRT so that each point on the object is reflected from the mirror into a spatial position corresponding to the distance of that point from a specified viewing position.

This allows a person to walk around an object or science and view it from different sides.

Real – time example

- Geisco space graph system – used in medical applications
- For analyzing data from ultrasonography & CAT scan devices.

- In geological applications to analyze topographical & scismic data.

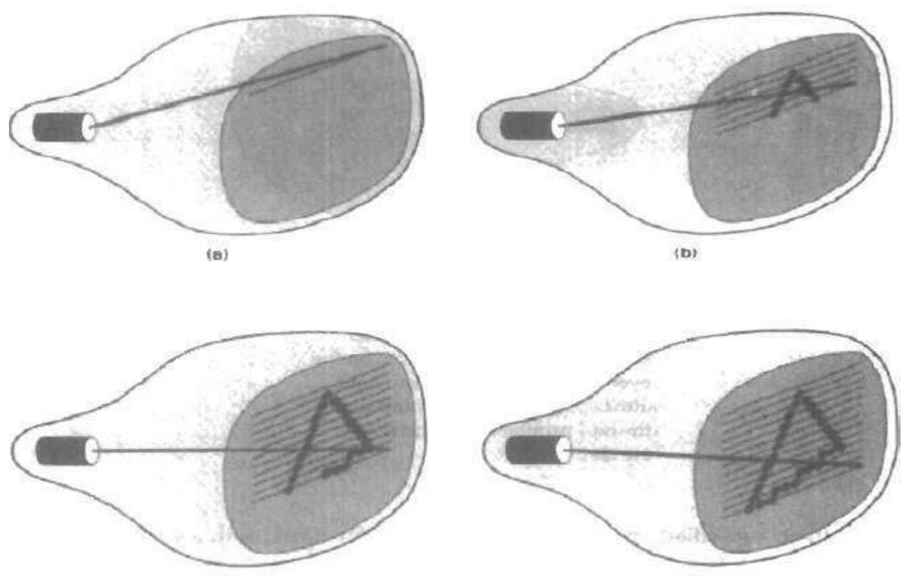
2. Discuss the functions of raster-scan and random scan displays.(nov (2012)

Raster Scan Displays

The most common type of graphics monitor employing a CRT is the Raster scan display.

Working Principle

- In a raster scan system the electron beam is swept across the screen one row at a time from top to bottom.
- When the electron beam moves across each row the beam intensity is turned on and off to create a pattern of illuminated spots.



- Picture definition is stored in memory area called Refresh buffer (or) Frame buffer.
- The frame buffer holds the set of intensity values for all the screen points.
- The stored intensity values are then retrieved from the refresh buffer and painted on the screen one row at a time.

- One row is also referred to as Scan Line.
- Each screen point is referred to as Pixel

Bit per Pixel

- Intensity range for pixel positions depend on the capability of the raster system.
- In black & white system each screen point is either on (or) off.

- Only one bit per pixel is needed to control the intensity of screen positions.

1 – electron beam is turned on

0 – electron beam is turned off

- High quality systems use upto 24 bits per pixel. The frame buffer of such system require several MB's of storage for the frame buffer.

Refreshing

- Refreshing of Raster scan displays is carried out as the rate of 60 to 80 frames per second.
- Refresh rates are described in units of cycles per second or Hertz.

  1 cycle – 1 frame

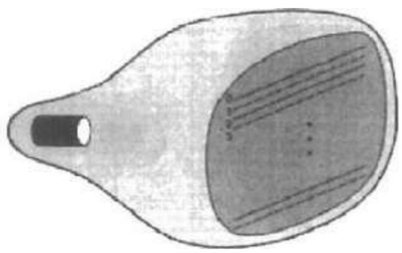
- At the end of each scan line the electron beam returns to the left side of the screen to begin displaying the next scan line.
- The return of the electron beam to the left of the screen is called the horizontal retrace.
- At the end of each frame the electron beam returns to the top left corner of the screen.
- This is referred to as vertical retrace.

- **Random – Scan Systems**

- In a Random- scan display unit a CRT has the electron beam directed only to the parts of the screen where a picture is to be drawn.

- Random scan monitors draw a picture one line at a time & for this reason they are referred to as vector displays.

The component lines of a picture can be drawn and refreshed by as random scan system in any specified order.



Refresh Rate

- Refresh rate depends on the number of lines to be displayed.
- A refresh buffer is used to store picture definition. The line drawing commands are stored in the refresh buffer.
- To display a specified picture the system processes the set of drawing commands.
- Random scan displays are designed to draw all the components lines of a picture 30 to 60 times each second.
- Random scan system are designed for line drawing applications and cannot display realistic shaded scenes.
- Random scan displays produce smooth line drawings because the CRT beam directly follows the line path.

3. List out and discuss the input devices. (apr 2013)(apr/may 2012)

Input Devices

- Various devices are available for data input on graphics workstations.

- Most systems have a keyboard & one or more additional devices specially designed for interactive input.

The various input devices that are discussed in this section are

xi) keyboards
xii) mouse
xiii) trackball & space ball
xiv) joysticks
xv) data glove
xvi) digitizers
xvii) image scanners
xviii) touch panels
xix) light pens
xx) voice systems

KEYBOARDS

- The keyboard is an efficient device for inputting such nongraphic data as picture labels associated with a graphic display.
- Alphanumeric keyboard

- Keyboards are provided with features to facilitate entry of screen co-ordinates, menu selections or graphics functions.

The common features on general purpose keyboards are

iii) cursor-control keys
iv) function keys

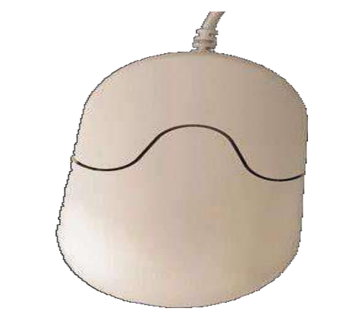
(i) Cursor control keys:

Can be used to select displayed objects or co-ordinate positions by positioning the screen cursor.

(ii)Functional keys:

Allow users to enter frequently used operations in a single keystroke. Additionally a numeric keypad is often included on the keyboard for fast entry of numeric data.

<u>MOUSE</u>



- A mouse is a small hand held device used to position the screen cursor.
- Wheels or rollers on the bottom of the mouse can be used to record the amount & direction of movement.
- Another method for detecting mouse movements is with an optical sensor.

- One, two or three buttons are usually included on the top of the mouse for signaling the execution of some operation.

Z-Mouse

- Additional devices can be included in the basic mouse design to increase the number of allowable input parameters.
- The z-mouse includes three buttons, a thumbwheel on the side, a trackball on the top & a standard mouse ball underneath.
- With the z-mouse, one can pick up an object, rotate it and move it in any direction or the navigation of one's viewing position & ordination through a 3D scene.
- Application of z-mouse are
- virtual reality
- CAD
- Animation
- Trackball And Spaceball

Trackball

- Trackball is a ball that can be rotated with the fingers or palm of the hand to produce screen cursor movements.
- Potentiometers attached to the ball measure the amount & direction of rotation.
- They are often mounted on keyboard or z-mouse.

Spaceball

- A space ball provides size degrees of freedom.
- A space ball does not actually move. Strain gauges measure the amount of pressure applied to the space ball.

Applications where space balls are used are

vi) 3D positioning

vii) Virtual reality systems
viii) Modeling
ix) Animation
x) CAD

Joysticks

A joystick consists of a small vertical lever called the stick

The stick is mounted on a base which steers the screen around.

There are 2 types of joystick

iii) movable stick joystick
iv) non-movable stick joystick

(i) Movable Stick Joystick

- The stick is actually placed in the center position.
- Screen cursor movement is achieved by moving the stick in any direction from the center.
- Potentiometers also mounted at the base of the joystick & they measure the amount of movement.
- Potentiometers also return the stick to the center position when released.
- In another type of movable joystick the stick is used to activate switches that cause the screen cursor to move at a constant rate in the selected direction.

Non Movable Stick Joystick

These joysticks are also called isometric joysticks.

They have a non movable stick & pressure applied on the joystick is measured by strain gauges & converted to the movement of the cursor in the specified direction.

Data Glove

- Data glove can be used to grasp a virtual object
- The glove has a series of sensors that detect hand & finger movements.
- Electromagnetic coupling between transmitting antennas & receiving antennas is used ot provide information about the position & orientation of the screen.
- Input from the glove can be used to position or manipulate objects in a virtual scene.
- A 2D projection of the scene can be viewed using video monitor.
- A 3D projection can be viewed with a headset.

Digitizers

- A digitizer is a common device for drawing, painting or interactively selecting co-ordinating positions on an object.
- Digitizers are used to input co-ordinate values in either a 2D or 3D space.
- A digitizer is used to scan over a drawing or object & to input a set of discrete co-ordinate positions.
- Graphics tablet is an example of digitizer.

Image Scanners

- An image scanner is used for storing drawings, graph, color & black and white photos or text by passing an optical scanning mechanism for computer processing.
- The gradations of gray scale or color are then recorded and stored in an array.
- Transformations can be applied to rotate, scale or crop the picture to a particular screen area.
- Various image processing methods can be applied to modify the array representations or text & they come in a variety of sizes & capabilities.

Touch Panels

- Touch panel allow displayed objects or screen positions to be selected with the touch of a finger.
- A typical application of touch panel's is for the selection of processing options that are represented with graphical icons.
- Eg ATM center touch panel
- Touch input can be recorded using optical, electrical or acoustical methods.

Optical Touch Panels

- These panels employ a line of infrared light emitting diodes (LED's) along one vertical edge & along one horizontal edge of the frame.
- The opposite vertical edge & horizontal edge contain light detectors.
- When the panel is touched these light detectors record which beams are interrupted.
- The 2 cross beams that are interrupted identify the horizontal & vertical co-ordinates of the screen position selected.
- Positions can be selected with an accuracy of about ¼ inch.
- The LED's operate at infrared frequencies so that the light is not visible to a user.

Electrical Touch Panel

- These plates are constructed with 2 transparent plates separated by a small distance.
- One plate is coated with a conducting material & the other with resistive material.
- Touching the outer plate forces it into contact with the inner plate.
- This contact creates a voltage drop across the resistive plate that is converted into co-ordinate values of the selected screen position.

Acoustical Touch Panel

- Here high frequency sound waves are generated by the horizontal & vertical directions across a glass plate.
- Touching the screen causes part of each wave to be reflected from the fingers to the emitters.

- The screen position is calculated from a measurement of the time interval between the transmission of each wave & its reflection to the emitter.

Light Pen

- Light pens are pencil-shaped devices used to select screen positions by detecting the light coming from points on the CRT screen.
- Light pens are sensitive to short burst of light emitted from the phosphor coating of the CRT.
- Other light sources are not usually detected by a light pen.
- An activated light pen pointed at a spot on the screen as the electron beam lights up that spot generates an electric pulse that causes the co-ordinate position of the electron beam to be recorded.
- The recorded light pen co-ordinates can be used to position an object or to select a processing option.

Disadvantages

v) Screen image obscured by hand & light pen

vi) Prolonged use causes arm fatigue

vii) Light pens require special implementation for some applications.

viii) Sometimes light pens give false readings due to background lighting in room.

Voice Systems

- Speech recognizers are used in some graphics workstations as input devices to accept voice commands.
- The voice system input can be used to initiate graphics operations or to enter data.
- These systems operate by matching an input against a predefined dictionary of words and pharses.

4.List out and discuss the output devices(apr/may 2012)

Output Devices

Speakers

The primary method of sound output in most computers today. Sound is translated from bits to electrical signals in a sound card, which then channels the signals to the speakers.

MIDI

- MIDI (Musical Instrument Digital Interface), is an industry-standard protocol that enables electronic musical instruments (synthesizers, drum machines), computers and other electronic equipment (MIDI controllers, sound cards, samplers) to communicate and synchronize with each other.
- Unlike analog devices, MIDI does not transmit an audio signal — it sends event messages about pitch and intensity, control signals for parameters such as volume, vibrato and panning, cues, and clock signals to set the tempo.
- MIDI files are typically created using computer-based sequencing software (or sometimes a hardware-based MIDI instrument or workstation) that organizes MIDI messages into one or more parallel "tracks" for independent recording and editing.

Microphone

A microphone is a device used to change sound into electric signals. Microphones are used in telephones, tape recorders, hearing aids and many other devices.

Printer

A printer is a device that prints text or images, making a physical copy (usually with some kind of ink on paper). Printers are classified into two types: Impact and non-impact printers.

(i) Impact printers rely upon a mechanical impact to transfer ink to paper. Early printers were more like electric typewriters, striking an ink ribbon against the paper with a lever with a raised image of a letter on the end, or a rotating ball with the same.

(ii) Non-impact printers depend on other ways of creating an image. Plotters, covered separately, can also be used as printers.

Laser

- A laser printer directs a laser beam onto a rotating drum, covered by a photoconductor (a material that conducts electricity when illuminated), carrying an electrostatic charge.
- The laser "erases" the charge in areas it strikes.
- Then a powdered ink ("toner"), charged the same polarity as the original surface charge on the drum, is spread across the surface, and it is repelled from the areas that still carry the original charge, but it is attracted to the discharged areas where the image was "written" by the laser.
- The drum then rolls the image formed onto paper, which is then be heated ("fused") to make the toner stick to itself and the paper.

Inkjet

An inkjet printer is a type of computer printer that creates a digital image by propelling variably-sized droplets of liquid material (ink) onto a page. Inkjet printers are the most common type of printer and range from small inexpensive consumer models to very large and expensive professional machines.

Hard-Copy Devices

- Hard copy output for images can be obtained in several formats
- Users can put pictures on paper by directing graphics output to a printer or plotter.
- The quality of pictures obtained from a device depends on the dot size & dot per inch or lines per inch.

- Smooth characters can be produced in printed text strings by high quality printers.
- These printers shift dot positions so that adjacent dots overlap

There are 2 major methods or types of printers namely

i) Impact devices &
ii) Non impact devices

(i) Impact Devices

- Impact printers press formed character faces against an inked ribbon onto paper
- A typeface is used to make the impression

Typefaces are normally mounted on bands, chains, drums or wheels.

Eg dot matrix printer

Dot Matrix Printer

- It is an example of impact printer
- Dot matrix printers have a dot matrix print head containing a rectangular array of protruding pins
- The no. of pins depends on the quality of the printer
- Individual characters or graphic patterns are obtained by retracting certain pins so that the remaining pins form the pattern to be printed

(ii) Non Impact Devices

Non impact plotters and printers use laser techniques, ink jet sprays, xerographic processes, electrostatic methods & electro thermal methods to get images on the paper

Laser Printer

- In a laser device , a laser beam creates a charge distribution on a rotating drum coated with a photoelectric material such as selerium
- Toner is applied to the drum and then transferred to paper
- The quality of the printout depends upon the dots per inch (dpi)

Ink Jet Printer

- These printers produce output by squirting ink in horizontal rows across a roll of paper wrapped on a drum.
- The electrically charged ink stream is deflected by an electric field to produce dot matrix patterns.

Electrostatic Printer

- These printers place a negative charge on the paper one complete row at a time along the length of the paper.
- The paper is then exposed to the toner
- The toner is positively charged and is attracted to negative charged areas on the paper.

Color Output On Impact And Non Impact Printers

- Limited color output can be obtained using non impact printers by using different color ribbons
- Various techniques are used by non impact printers to combine three color pigments (cyan, magenta & yellow)
- Laser & xerographic devices deposit the three pigments on separate passes
- Ink jet methods shoot the three colors simultaneously on a single pass along each print line on the paper

5. Explain the function of following input devices (a)Digitizer (b)joystick (c)Imag e

Scanner(nov 2012)

Digitizers

- A digitizer is a common device for drawing, painting or interactively selecting co-ordinating positions on an object.
- Digitizers are used to input co-ordinate values in either a 2D or 3D space.
- A digitizer is used to scan over a drawing or object & to input a set of discrete co-ordinate positions.
- Graphics tablet is an example of digitizer.

Joysticks

A joystick consists of a small vertical lever called the stick

The stick is mounted on a base which steers the screen around.

There are 2 types of joystick

v) movable stick joystick

vi) non-movable stick joystick

(ii) Movable Stick Joystick

- The stick is actually placed in the center position.
- Screen cursor movement is achieved by moving the stick in any direction from the center.
- Potentiometers also mounted at the base of the joystick & they measure the amount of movement.
- Potentiometers also return the stick to the center position when released.

- In another type of movable joystick the stick is used to activate switches that cause the screen cursor to move at a constant rate in the selected direction.

Non Movable Stick Joystick

These joysticks are also called isometric joysticks.

They have a non movable stick & pressure applied on the joystick is measured by strain gauges & converted to the movement of the cursor in the specified direction.

Image Scanners

- An image scanner is used for storing drawings, graph, color & black and white photos or text by passing an optical scanning mechanism for computer processing.
- The gradations of gray scale or color are then recorded and stored in an array.
- Transformations can be applied to rotate, scale or crop the picture to a particular screen area.
- Various image processing methods can be applied to modify the array representations or text & they come in a variety of sizes & capabilities.

6.Explain the interactive picture construction techniques in details(apr/may 2012)

Interactive Picture construction techniques

An interactive construction technique means we will use a technique to construct a picture. There are different picture construction techniques are available to construct a picture.

(i) **Positioning** – In this method we will position different objects according to the requirement of an application. We will use various input device like mouse and various other device to change the location of an object. In this method we will decide the position of an object.

Advantage:

- We can easily change the location of an object with the help of mouse
- We can easily view where our object will appear on the window.
- We can easily observe one object can overlap with each other or not.

Disadvantage

- We cannot get the actual position of an object.
- We cannot calculate accurate positioning point of an objects.

(ii) Dragging - In this method we will drag the object from one location to another location .In this method we will select an object from one location and drag that object to another location. This method is quite good to see the appearance of an object by dragging them from one location to another location.

Advantage:

- We can easily drag an object from one location to another location
- We can easily see the appearance of final output that will come in the front of us.

Disadvantage

- We are totally bounded to mouse.
- Floating point position cannot be given to change the location of an object

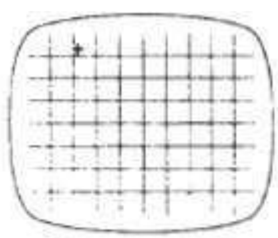
(iii) Constraints - In this method we will use various rules that can be used to make a particular picture. Constraints will contain various rules in the form of tools. Constraints can be implemented according to the requirement of the user. Constraints can include eraser and shape of pens and different other options.

- **Advantage:**
- We can modify a picture according to our requirement.
- We can perform various operation on picture modification.
- We can easily implement the rules over picture construction and user can be bounded to do the task.

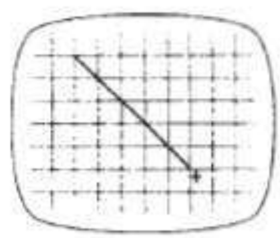
Disadvantage

Some time one rule or constraint cannot be implemented on any application or object.

(iv) Grids -In this method we will divide a picture into grids. Grids will be constructed according to the size of an image. If image is large then grid will be more otherwise grid will be less. Object will be arranged according to intersection of row and column . Where intersection is happening where object will be placed. Suppose one object location position at 4.6 then it is automatically shifted to 5 if it is at 4.3 then it is automatically shifted to 4.



Select First Endpoint Position Near a Grid Intersection



Select a Position Near a Second Grid Intersection

Advantage:

- Object will provide accurate position of an object
- Object can be moved from one grid location to another location
- Object are automatically shifted to rounded location

Disadvantage:

- We cannot get accurate location of an object.
- We cannot get accurate value of an object.

his method we will make gravity field around a particular line. Gravity field will make a gravity around a line that look like any two lines are combining with each other.

Advantage:

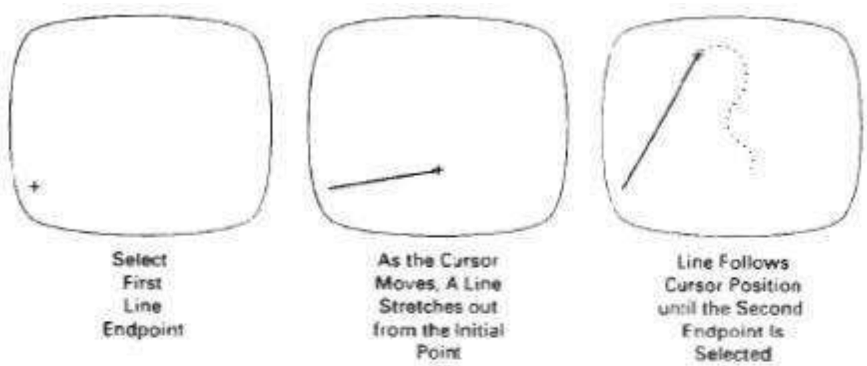
- We can combine various shapes with each other without combining them.
- We can change the appearance of different shapes without any changes in actual shape.

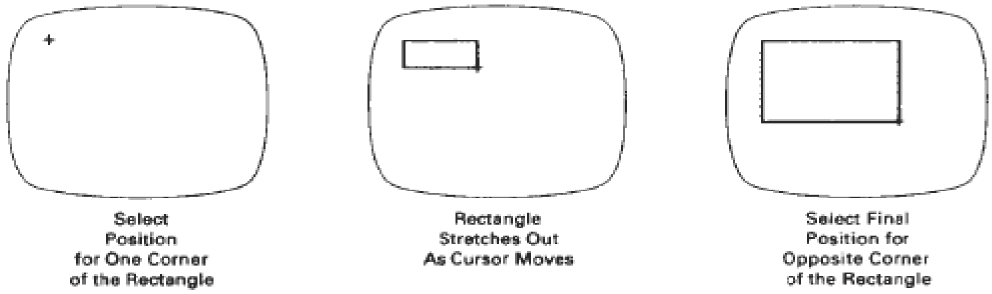
Disadvantage:

- We cannot get actual appearance of a picture.
- We cannot get the actual co-ordinates of an object.

(vi) Rubber-Band Method.

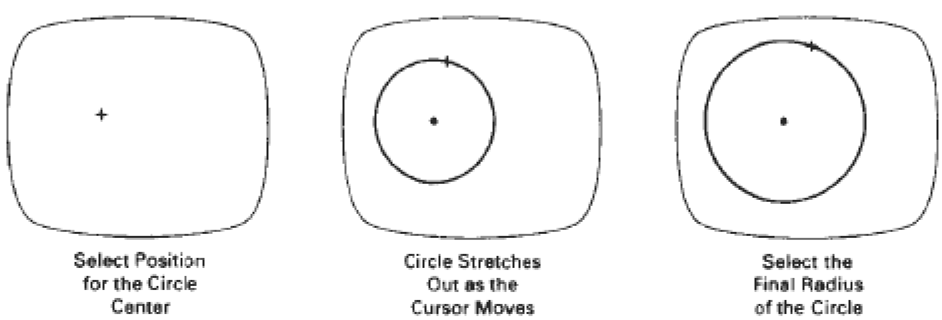
- Straight lines can be constructed and positioned using ***rubber-band*** methods, which stretch out a line from a starting position as the screen cursor is moved. Rubber-band methods are used to construct and position other objects besides straight lines.





(vii) Painting and Drawing

- Options for sketching, drawing, and painting come in a variety of forms. Straight
- lines, polygons, and circles can be generated with methods
- Curve drawing options can be provided using standard curve shapes, such as circular arcs and splines, or with freehand sketching procedures.
- Splines are interactively constructed by specifying a set of discrete screen points that give the general shape of the curve.
- Then the system fits the set of points with a polynomial curve. In freehand drawing, curves are generated by following the path of a stylus on a graphics tablet or the path of the screen cursor on a video monitor. Once a curve is displayed, the designer can alter the curve shape
- By adjusting the positions of selected points along the curve path.



- Line widths, line styles, and other attribute options are also commonly found in -painting and drawing packages.